

Adaptive Routing under Mobility and Failures

A Reproducible ns-3 Pilot with Robustness and Component Ablation

Wisam Makki Salim

Computer Engineering Researcher | Al-Iraqia University, Baghdad, Iraq
ORCID 0009-0000-6998-3912

Abstract

This pilot tests whether lightweight OLSR timer adaptation can preserve service under mobility, link degradation, and node failure. A reproducible ns-3.47 experiment compares standard OLSR with a controller that reacts to weak signals and final transmission failures. The evaluation uses a 16-node wireless ad hoc topology, two UDP flows, three controlled scenarios, and 20 paired seeds per comparison. The locked evaluation does not support the original superiority hypothesis: service effects vary across seeds while routing overhead rises whenever adaptation triggers. Component ablation isolates accelerated HELLO messaging as the source of an adverse cross-flow tail. A lower failure threshold yields exploratory gains but remains unconfirmed. The contribution is executable evidence about when a plausible adaptive action fails, why it fails in the tested configuration, and how to redesign the question for doctoral research.

Research Question and Hypothesis

Can a lightweight adaptive routing-control mechanism improve service resilience under changing mobility, link degradation, and failure conditions compared with a static baseline, without introducing excessive latency or routing overhead?

The working hypothesis predicted shorter disruption and recovery with a modest control cost. The experiment tests this claim rather than assuming that faster protocol control is beneficial.

Experimental Design

Element	Locked specification
Simulator	ns-3.47; IEEE 802.11g ad hoc; standard OLSR baseline
Network	16-node jittered 4 by 4 grid; two concurrent UDP flows; 512-byte packets every 20 ms
Adaptation	EWMA link signal and final transmission failures; reactive HELLO 0.5 s and TC 1.0 s
Scenarios	A normal operation; B gradual next-hop displacement; C active next-hop radio failure
Evaluation	Runs 101 to 120 paired by seed; 50,000 deterministic bootstrap resamples
Measures	PDR, worst-flow PDR, delay, goodput, control overhead, recovery, and route stability

The controller enters a reactive state after three weak-signal samples at or below -81 dBm or 40 final transmission failures within one second. Hysteresis and a ten-second hold-down govern reversion. Runs are retained regardless of direction or magnitude.

Locked Evaluation Results

Scenario A acts as a negative control: baseline and adaptive outputs are identical in all 20 pairs and no trigger fires. In Scenario B, the PDR interval crosses zero while normalized overhead increases in every pair. In Scenario C, the median PDR effect is zero, but run 102 reveals a severe adverse tail that is reproduced exactly with diagnostic tracing.

Case	Median PDR baseline	Median PDR adaptive	Mean paired PDR effect 95 percent CI	Evidence
A	100.00%	100.00%	0.000 [0.000, 0.000]	No trigger; identical outputs
B	97.10%	97.06%	+0.057 [-0.119, +0.312]	No stable benefit; overhead higher in 20 of 20
C	97.07%	97.21%	-0.659 [-2.777, +0.604]	Typical effect near zero; adverse tail retained

Paired PDR differences across final-evaluation seeds

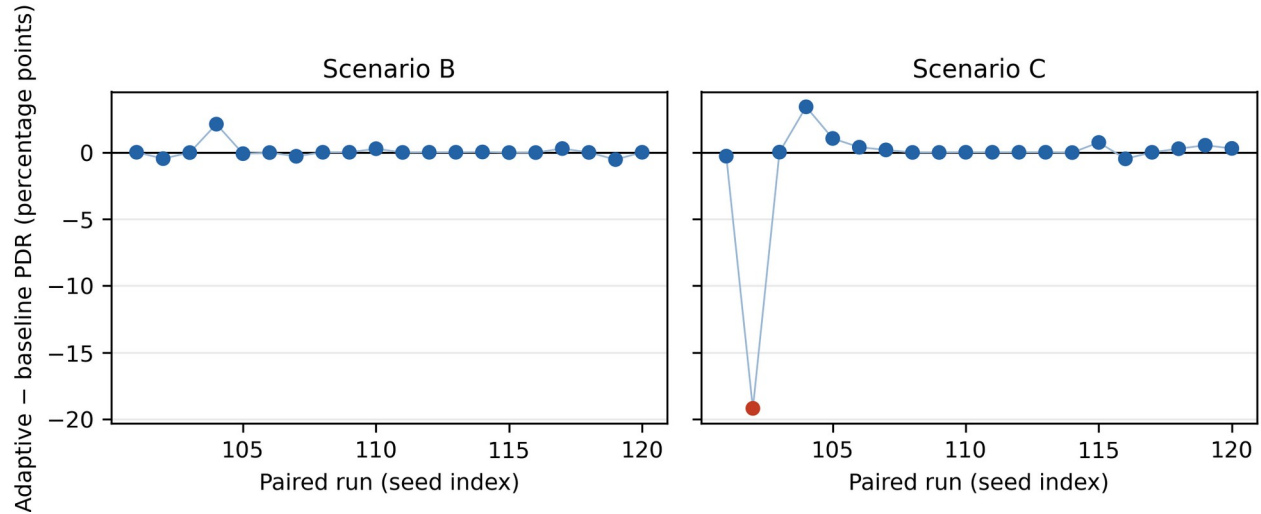


Figure 1 Adaptive minus baseline PDR for every paired final evaluation run

Interpretation

The Scenario B mean effect is +0.057 percentage points with a 95 percent bootstrap interval from -0.119 to +0.312. Adaptive PDR is lower in 10 pairs, equal in 6, and higher in 4. The mean overhead difference is +0.002518 with an interval wholly above zero. The mechanism therefore incurs a repeatable cost without a repeatable service benefit under gradual degradation.

In Scenario C, 9 pairs improve, 4 are equal, and 7 decline. Run 102 reduces PDR from 95.51 percent to 76.34 percent and extends recovery from 7 s to 36 s. The rerun matches the stored result exactly. Per-flow tracing shows that one flow recovers while another remains largely undelivered despite a source route.

Component Ablation

Scenario C was rerun with detection-only, HELLO-only, and topology-control-only variants. Detection-only matches baseline in every network field across all 20 runs. HELLO-only matches the full adaptive mechanism exactly. Topology-control-only is nearly identical to baseline. The evidence therefore attributes the measured cost and adverse tail to shortening HELLO from 2.0 s to 0.5 s.

Scenario C component ablation: paired mean effects with 95% bootstrap CI

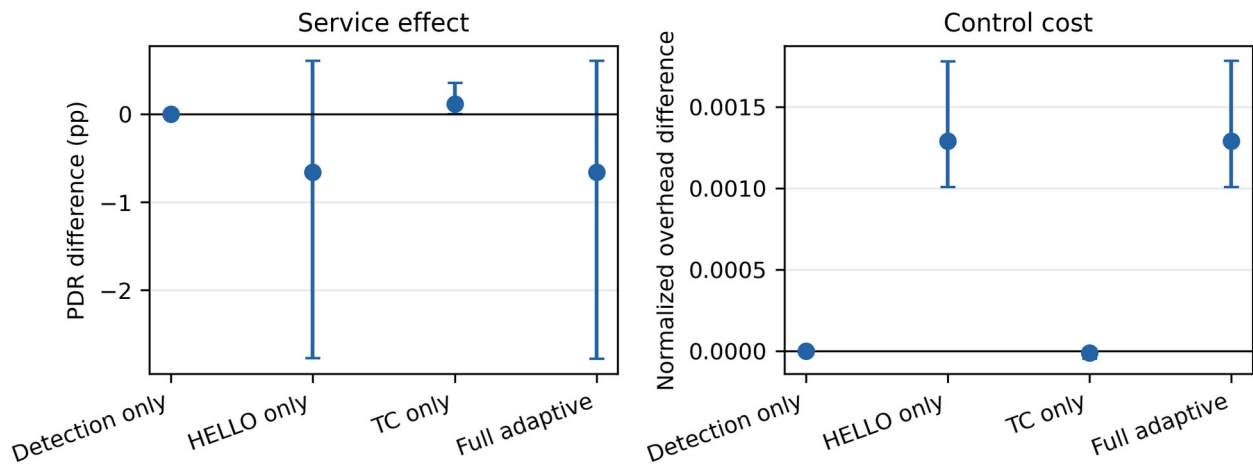


Figure 2 Scenario C component ablation with paired mean effects and bootstrap intervals

Run 102 variant	PDR	Overhead	Recovery	Result
Baseline	95.51%	0.017424	7 s	Reference
Detection only	95.51%	0.017424	7 s	Exact baseline match
HELLO only	76.34%	0.023060	36 s	Exact full-adaptive match
TC only	95.51%	0.017424	7 s	Exact baseline match
Full adaptive	76.34%	0.023060	36 s	Adverse tail

Causal Boundary

The ablation provides strong within-simulation evidence that accelerated HELLO messaging creates the run 102 outcome. It does not establish the lower-level cause. The available traces cannot distinguish transient routing-state inconsistency from wireless contention or another OLSR interaction. That distinction requires additional instrumentation rather than inference from aggregate delivery.

Threshold Sensitivity

Threshold sensitivity: paired mean PDR effects with 95% bootstrap CI

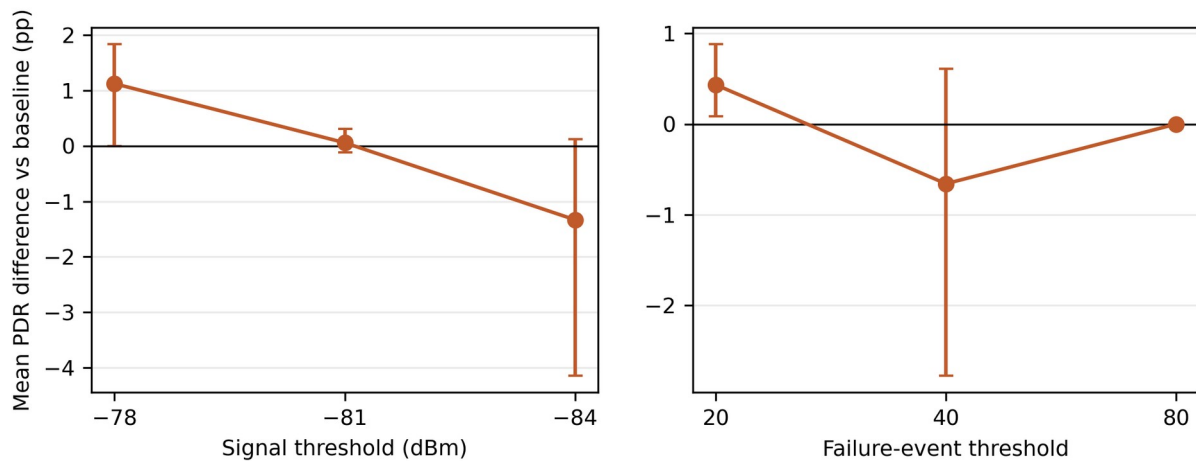


Figure 3 Threshold sensitivity with paired mean PDR effects and bootstrap intervals

A failure threshold of 20 events produces a mean PDR gain of +0.434 percentage points [95 percent CI +0.082, +0.884] and a mean recovery reduction of 0.45 s. Its leave-one-out PDR mean remains positive, but overhead increases in every run. This is an exploratory redesign candidate, not a confirmed result, because it reuses the evaluation seeds and topology. A threshold of 80 never triggers and only reproduces baseline behavior.

In Scenario B, -78 dBm produces 317 signal transitions and a large overhead increase. At -84 dBm, signal triggering disappears and the failure channel triggers in 19 runs. The channels substitute for one another, so these data do not define an optimal RSSI threshold.

Limitations and Doctoral Direction

The pilot uses one topology family, one density, two constant-rate flows, and an idealized failure. It does not measure channel occupancy or route loops directly. Aggregate recovery can conceal flow-level unfairness, and bootstrap intervals over 20 paired runs do not establish generality across network environments.

A doctoral extension should replace unconditional timer acceleration with guarded control that combines link risk, route state, and congestion evidence; limits the duration and scope of HELLO changes; and treats cross-flow harm as a first-class outcome. Confirmation requires fresh seeds, multiple mobility and topology families, heterogeneous traffic, and explicit tail-risk and fairness measures. Adversarial disruption can then be introduced as a separate controlled factor.

Conclusion

The original timer-switching mechanism does not provide reliable evidence of improved resilience. It raises overhead and can create severe cross-flow harm after failure. Although the original mechanism was not supported, the pilot establishes a reproducible experimental basis for investigating guarded adaptive routing under failure. Its controlled baselines, uncertainty analysis, diagnostic traces, and component ablation define a focused doctoral research direction.

References

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